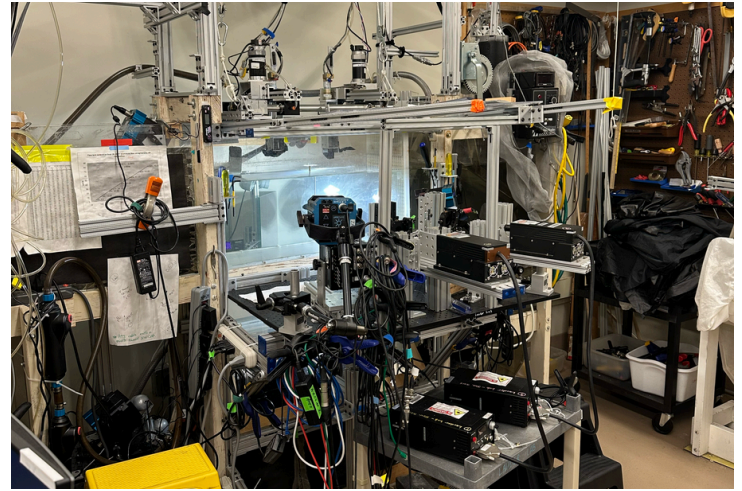


FISH ROBOT SWIMMING EFFICIENCY - HARVARD UNIVERSITY



What?

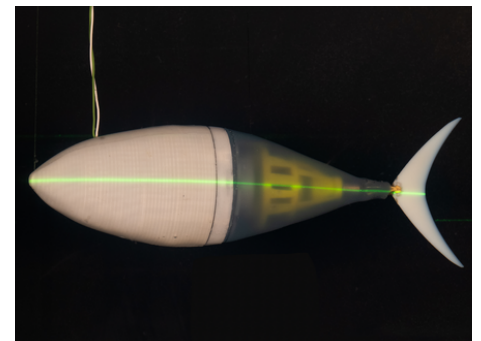
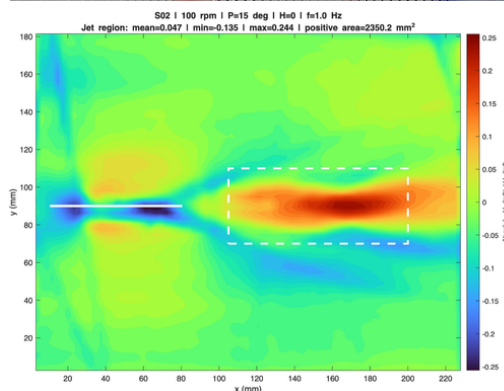
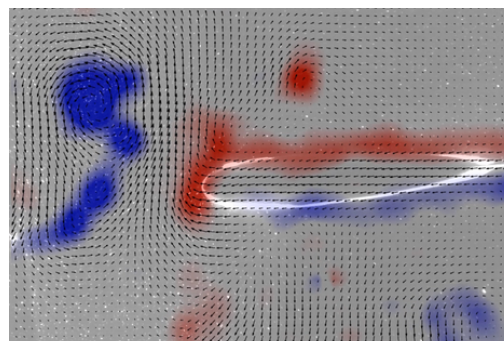
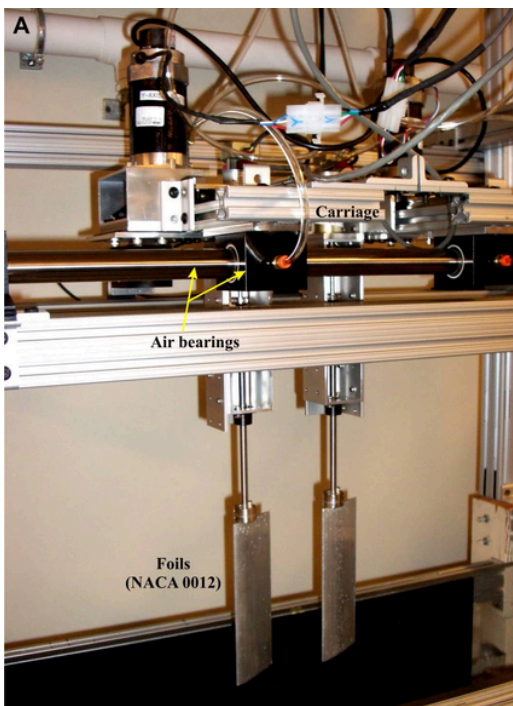
- Studied how a fish swimming in line behind a leading swimmer gains thrust from its wake without actively swimming, with applications to ships and underwater drones.
- Measured how frequency, pitch amplitude, and flow speed affect swimming and station-holding performance.

How?

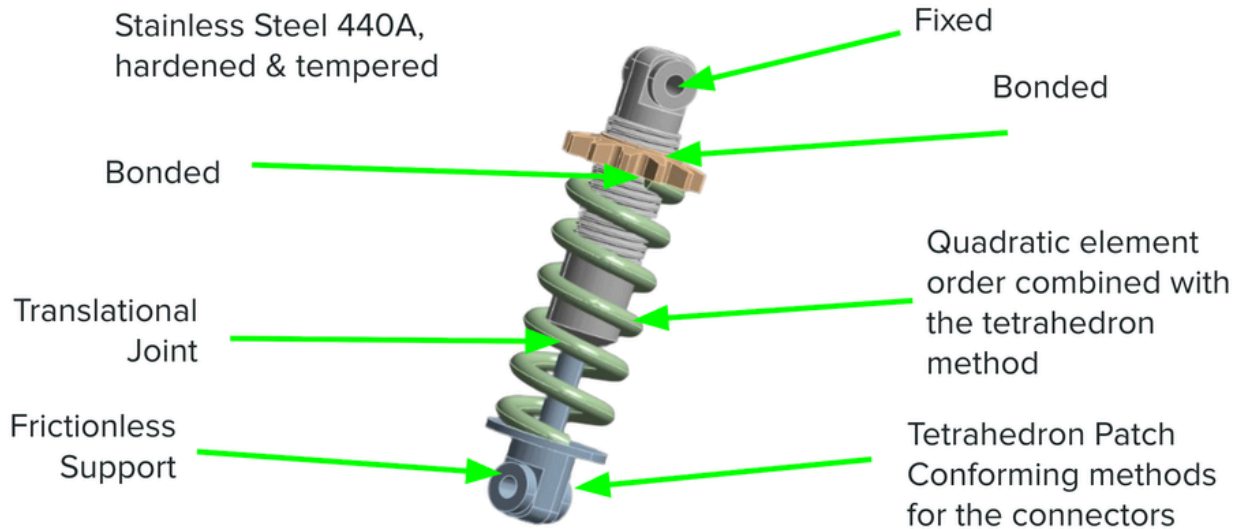
- Operated and redesigned a robotic flapper with air bearings, LabVIEW, PID control, and NI-DAQ.
- Ran hydrofoil and passive-fish experiments across flow conditions.
- Performed particle image velocimetry (PIV); processed wake data in MATLAB and PIVlab.
- Built 2-D tracking and 3-D reconstruction pipelines in MATLAB and R.

Results

- Presented findings at the ONR MURI Annual Review Meeting (Sep 2026) and the APS Division of Fluid Dynamics Meeting (Nov 2026).
- Quantified swimming kinematics across flow conditions, which can inform ship propulsion and underwater drone design.



SHOCK ABSORBER - FEA MODAL ANALYSIS



What?

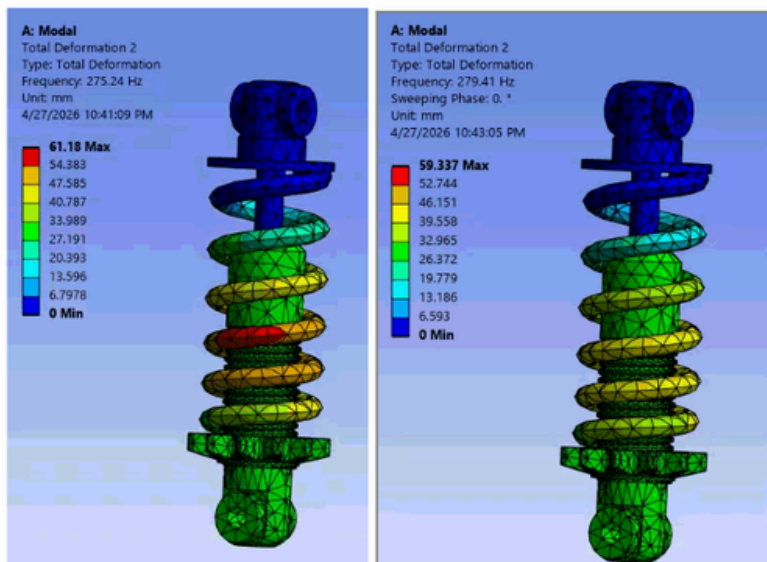
- Performed modal analysis on a scaled shock absorber assembly to study its natural vibration behavior.
- Tested how damping affects vibration decay and stability.

How?

- Set up the model in ANSYS with fixed, bonded, frictionless, and translational joint boundary conditions.
- Used Stainless Steel 440A and quadratic tetrahedral mesh elements.
- Tested damping ratios from 0.05 to 0.3.

Results

- Natural frequencies stayed around 152–712 Hz.
- Higher damping caused faster vibration decay without major frequency shifts.
- Results showed the system is mainly mass–stiffness dominated.
- Fixed contact issues by improving boundary conditions and using a translational joint where sliding motion was needed.



Damping off

Damping on

0.05 damping ratio

Mode	Damped Frequency [Hz]	Stability [Hz]	Modal Damping Ratio	Logarithmic Decrement
1. 1.	151.89	-1.551	1.0211e-002	-6.4161e-002
2. 2.	355.23	-5.3157	1.4963e-002	-9.4023e-002
3. 3.	367.36	-5.5009	1.4994e-002	-9.4222e-002
4. 4.	414.7	-6.1958	1.4939e-002	-9.3874e-002
5. 5.	419.73	-6.2565	1.4904e-002	-9.3658e-002
6. 6.	704.77	-10.564	1.4988e-002	-9.4183e-002

0.1 damping ratio

Mode	Damped Frequency [Hz]	Stability [Hz]	Modal Damping Ratio	Logarithmic Decrement
1. 1.	152.02	-5.1625	3.3939e-002	-0.21337
2. 2.	355.63	-17.699	4.9706e-002	-0.3127
3. 3.	367.77	-18.342	4.9811e-002	-0.31336
4. 4.	415.17	-20.629	4.9627e-002	-0.3122
5. 5.	420.2	-20.832	4.9515e-002	-0.31149
6. 6.	705.59	-35.176	4.9791e-002	-0.31323

0.2 damping ratio

Mode	Damped Frequency [Hz]	Stability [Hz]	Modal Damping Ratio	Logarithmic Decrement
1. 1.	152.46	-10.276	6.7251e-002	-0.42351
2. 2.	356.94	-35.268	9.8328e-002	-0.62082
3. 3.	369.13	-36.549	9.8533e-002	-0.62213
4. 4.	416.7	-41.106	9.817e-002	-0.61981
5. 5.	421.73	-41.512	9.7959e-002	-0.61847
6. 6.	708.23	-70.097	9.8482e-002	-0.62187

0.3 damping ratio

Mode	Damped Frequency [Hz]	Stability [Hz]	Modal Damping Ratio	Logarithmic Decrement
1. 1.	153.16	-15.297	9.938e-002	-0.62753
2. 2.	359.06	-52.587	0.14491	-0.92021
3. 3.	371.34	-54.498	0.14521	-0.92214
4. 4.	419.19	-61.231	0.14467	-0.91867
5. 5.	424.22	-61.901	0.14439	-0.91682
6. 6.	712.5	-104.52	0.14515	-0.92175

ABYLAIKHAN MUKHAMEJANOV

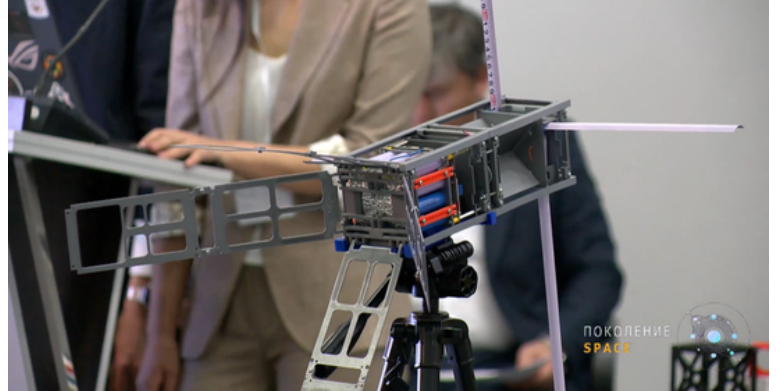
MECHANICAL ENGINEERING AT LEHIGH UNIVERSITY

✉ abylaikhan.mukhamejanov@gmail.com

in [linkedin.com/in/abylaikhan-m/](https://www.linkedin.com/in/abylaikhan-m/)

☎ +1 (484) 353-9865
abylaikhanm.github.io

SKILLSAT STARTUP



What?

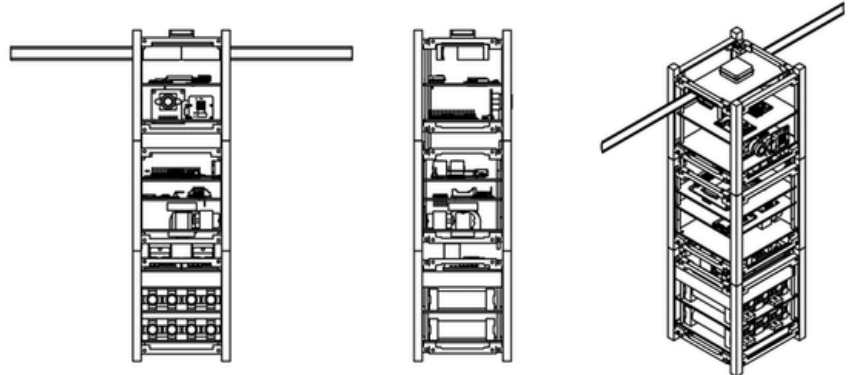
- Founded SkillSat, an EdTech startup to manufacture STEM kits that teach students to build stratospheric satellites.

How?

- Used Arduino and Raspberry Pi as the base for satellite systems.
- Utilized additive manufacturing to prototype 25+ designs.
- Applied DFA and DFM principles to reduce production costs by 44%.
- Tested prototypes at the National Space Center of Kazakhstan.

Results

- Launched satellites to Near Space (40 km+) via stratospheric balloons.
- Raised \$15K in funding and distributed 80+ kits to 240+ students across 25+ schools.
- Recognized as a Top-3 SpaceTech and Top-15 EdTech Startup by Forbes Kazakhstan.



ABYLAIKHAN MUKHAMEJANOV

MECHANICAL ENGINEERING AT LEHIGH UNIVERSITY

✉ abylaikhan.mukhamejanov@gmail.com

in [linkedin.com/in/abylaikhan-m/](https://www.linkedin.com/in/abylaikhan-m/)

☎ +1 (484) 353-9865
abylaikhanm.github.io



Uncovering the Fluid-Structure Interactions of Wave-Assisted Propulsion via Cyber-Physical Fluid Dynamics



Abylaikhan Mukhamejanov¹, Ata Tankut Ardic¹, Keith W. Moored¹
¹Department of Mechanical Engineering & Mechanics, Lehigh University, Bethlehem, PA 18015

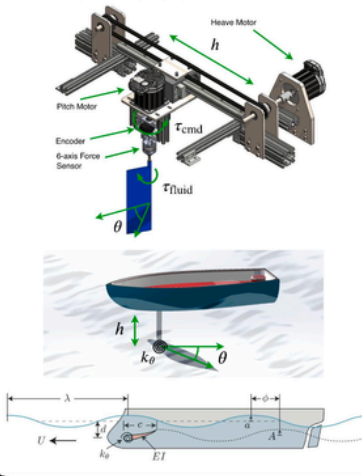
78th Annual Meeting APS DFD
Nov. 23–25, 2025, Houston, TX

Project Objectives and Goals

- Investigate wave-assisted propulsion (WAP) using controlled heave and passive pitch to transform wave energy into forward thrust and improve stability.
- Develop and validate a cyber-physical testbed that couples a flapping hydrofoil with virtual mass-spring-damper dynamics.
- Implement real-time control of virtual stiffness, damping, and inertia to emulate passive pitching behavior.

Motivation

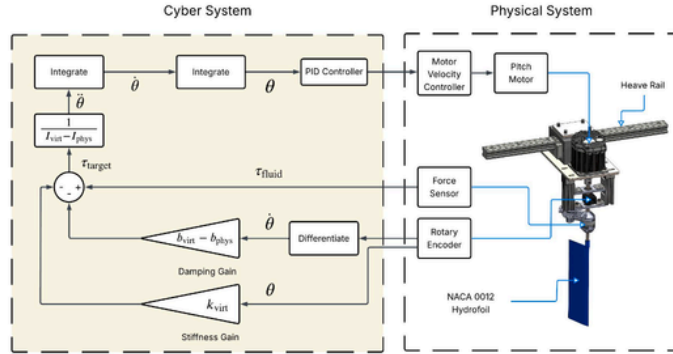
- Wave-assisted propulsion (WAP) uses oscillating hydrofoils to extract energy from waves and generate forward thrust.
- The flapping motion can also reduce vessel oscillations, improving stability and efficiency in marine systems.
- Studying these fluid-structure interactions (FSI) requires precise control of mass, stiffness, and damping parameters.
- Traditional experiments fix mass, stiffness, and damping through mechanical design, limiting flexibility in studying dynamic behavior.



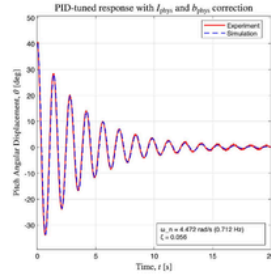
Methods & Procedures

- A cyber-physical system [1] couples the physical actuator-fluid testbed with a computer-based controller that enforces a virtual dynamic model in real time.
- Command signal cancels physical inertia and damping to match desired virtual parameters.
- Physical inertia and damping identified via ring-down experiment.
- Double integration [2] of force/torque removes filtering need and enables stable 4 kHz Simulink Real-Time control loop.

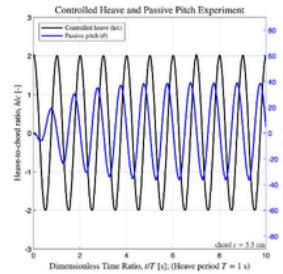
$$\begin{aligned}\tau &= I \ddot{\theta} + b \dot{\theta} + k \theta \\ \tau_{\text{virt}} &= -(I_v \ddot{\theta} + b_v \dot{\theta} + k_v \theta) \\ I_p \ddot{\theta} &= \tau_{\text{fluid}} + \tau_{\text{virt}} \\ (I_p + I_v) \ddot{\theta} + (b_p + b_v) \dot{\theta} + k_v \theta &= \tau_{\text{fluid}}\end{aligned}$$



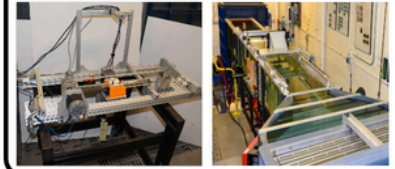
System Validation



Air Experiment Results



Experimental Setup



Conclusions

- Developed a working cyber-physical system mimicking passive pitching.
- Any structural parameters can be virtually assigned to the system, including nonlinear stiffness elements and damping characteristics.
- The system accepts real heave motion inputs from naval vessels, enabling experimental investigation and optimization of Wave-Assisted Propulsion (WAP).

Future Studies / Recommendations

- Evaluate propulsion efficiency and stability under varying environmental and structural parameters in the water channel.
- Recreate natural sea-state conditions by coupling heave-wave excitation with vessel motions across multiple degrees of freedom.

Acknowledgments

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References

- [1] Mackowski, A. W., and Williamson, C. H. K., 2011, "Developing a Cyber-Physical Fluid Dynamics Facility for Fluid-Structure Interaction Studies," *J. Fluids Struct.*, 27 (5–6), pp. 748–757.
- [2] Zhu, Y., Su, Y., and Breat, K., 2020, "Nonlinear Flow-Induced Instability of an Elastically Mounted Pitching Wing," *J. Fluid Mech.*, 899, A35.